

SYSTEMS OF LINEAR EQUATIONS

EXERCISE 4 (Page 174)

Solve the following systems of linear equations, the field of scalars being R :

$$\begin{aligned} 1. \quad & 2x_1 + x_3 = 1 \\ & 2x_1 + 4x_2 - x_3 = -2 \\ & x_1 - 8x_2 - 3x_3 = 2 \end{aligned}$$

$$\begin{aligned} \text{Sol.} \quad A_b &= \left[\begin{array}{ccc|c} 2 & 0 & 1 & 1 \\ 2 & 4 & -1 & -2 \\ 1 & -8 & -3 & 2 \end{array} \right] \\ &\stackrel{R}{\sim} \left[\begin{array}{ccc|c} 1 & -8 & -3 & 2 \\ 2 & 4 & -1 & -1 \\ 2 & 0 & 1 & 1 \end{array} \right], \text{ by } R_{13} \\ &\stackrel{R}{\sim} \left[\begin{array}{ccc|c} 1 & -8 & -3 & 2 \\ 0 & 20 & 5 & -6 \\ 0 & 16 & 7 & -3 \end{array} \right], \text{ by } R_2 - 2R_1 \text{ and } R_3 - 2R_1 \\ &\stackrel{R}{\sim} \left[\begin{array}{ccc|c} 1 & -8 & -3 & 2 \\ 0 & 4 & -2 & -3 \\ 0 & 16 & 7 & -3 \end{array} \right], \text{ by } R_2 - R_3 \end{aligned}$$

$$\underset{\sim}{R} \left[\begin{array}{ccc|c} 1 & -8 & -3 & 2 \\ 0 & 4 & -2 & -3 \\ 0 & 0 & 15 & 9 \end{array} \right], \text{ by } R_3 - 4R_2$$

$$\underset{\sim}{R} \left[\begin{array}{ccc|c} 1 & 0 & -7 & -4 \\ 0 & 4 & -2 & -3 \\ 0 & 0 & 1 & 3/5 \end{array} \right], \text{ by } R_1 + 2R_2 \text{ and } \frac{1}{15}R_3$$

$$\underset{\sim}{R} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1/5 \\ 0 & 4 & 0 & -9/5 \\ 0 & 0 & 1 & 3/5 \end{array} \right], \text{ by } R_1 + 7R_3 \text{ and } R_2 + 2R_3$$

$$\underset{\sim}{R} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1/5 \\ 0 & 1 & 0 & -9/20 \\ 0 & 0 & 1 & 3/5 \end{array} \right], \text{ by } \frac{1}{4}R_2$$

$$\text{Hence } x_1 = \frac{1}{5}, \quad x_2 = -\frac{9}{20}, \quad x_3 = \frac{3}{5}$$

$$\begin{aligned} 2. \quad & x_1 + \quad \quad x_2 + \quad \quad x_3 = a \\ & x_1 + (1+a)x_2 + \quad \quad x_3 = 2a, \quad a \neq 0 \\ & x_1 + \quad \quad x_2 + (1+a)x_3 = 3a \end{aligned}$$

$$\text{Sol. } A_b = \left[\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 1 & 1+a & 1 & 2a \\ 1 & 1 & 1+a & 3a \end{array} \right]$$

$$\underset{\sim}{R} \left[\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 0 & a & 0 & a \\ 0 & 0 & a & 2a \end{array} \right], \text{ by } R_2 - R_1 \text{ and } R_3 - R_1$$

$$\underset{\sim}{R} \left[\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right], \text{ by } \frac{1}{a}R_2 \text{ and } \frac{1}{2a}R_3$$

$$\underset{\sim}{R} \left[\begin{array}{ccc|c} 1 & 0 & 1 & a-1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right], \text{ by } R_1 - R_2$$

$$\underset{\sim}{R} \left[\begin{array}{ccc|c} 1 & 0 & 0 & a-3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right], \text{ by } R_1 - R_3$$

$$\text{Therefore } x_1 = a-3, \quad x_2 = 1, \quad x_3 = 2.$$

3.
$$\begin{aligned} x_1 - x_2 + x_3 - x_4 + x_5 &= 1 \\ 2x_1 + x_2 + 3x_3 + 4x_5 &= 3 \\ 3x_1 - 2x_2 + 2x_3 + x_4 + x_5 &= 1 \\ x_2 + x_4 + x_5 &= 0 \end{aligned}$$

Sol. $A_b = \left[\begin{array}{ccccc|c} 1 & -1 & 1 & -1 & 1 & 1 \\ 2 & 1 & 3 & 0 & 4 & 3 \\ 3 & -2 & 2 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \end{array} \right]$

$\tilde{R} \left[\begin{array}{ccccc|c} 1 & -1 & 1 & -1 & 1 & 1 \\ 0 & 3 & 1 & 2 & 2 & 1 \\ 0 & 1 & -1 & 4 & -2 & -2 \\ 0 & 1 & 0 & 1 & 1 & 0 \end{array} \right], \text{ by } R_2 - 2R_1 \text{ and } R_3 - 3R_1$

$\tilde{R} \left[\begin{array}{ccccc|c} 1 & -1 & 1 & -1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 4 & -2 & -2 \\ 0 & 3 & 1 & 2 & 2 & 1 \end{array} \right], \text{ by } R_{24}$

$\tilde{R} \left[\begin{array}{ccccc|c} 1 & 0 & 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & -1 & 3 & -3 & -2 \\ 0 & 0 & 1 & -1 & -1 & 1 \end{array} \right], \text{ by } R_1 + R_2, \\ R_3 - R_2 \text{ and } R_4 - 3R_2$

$\tilde{R} \left[\begin{array}{ccccc|c} 1 & 0 & 0 & 1 & 3 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 2 & -4 & -1 \\ 0 & 0 & 1 & -1 & -1 & 1 \end{array} \right], \text{ by } R_1 - R_4 \text{ and } R_3 + R_4$

$\tilde{R} \left[\begin{array}{ccccc|c} 1 & 0 & 0 & 1 & 3 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & 0 & 2 & -4 & -1 \end{array} \right], \text{ by } R_{34}$

$\tilde{R} \left[\begin{array}{ccccc|c} 1 & 0 & 0 & 1 & 3 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & 0 & 1 & -2 & -1/2 \end{array} \right], \text{ by } \frac{1}{2}R_4$

$$R \left[\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & 5 & 1/2 \\ 0 & 1 & 0 & 0 & 3 & 1/2 \\ 0 & 0 & 1 & 0 & -3 & 1/2 \\ 0 & 0 & 0 & 1 & -2 & -1/2 \end{array} \right], \quad \begin{array}{l} \text{by } R_1 - R_4, \\ R_2 - R_4 \text{ and} \\ R_3 + R_4 \end{array}$$

We see that $\text{rank } A = \text{rank } A_b = 4 < \text{the number of unknowns (which are 5)}$. Hence infinitely many solutions exist by assigning arbitrary value to any one of the unknowns, say x_5 . Thus we have

$$x_1 + 5x_5 = \frac{1}{2} \Rightarrow x_1 = \frac{1}{2} - 5x_5$$

$$x_2 + 3x_5 = \frac{1}{2} \Rightarrow x_2 = \frac{1}{2} - 3x_5$$

$$x_3 - 3x_5 = \frac{1}{2} \Rightarrow x_3 = \frac{1}{2} + 3x_5$$

$$x_4 - 2x_5 = -\frac{1}{2} \Rightarrow x_4 = -\frac{1}{2} + 2x_5$$

$$\begin{array}{rcl} 4. & x_1 + x_2 - x_3 & = 1 \\ & x_2 + x_3 - x_4 & = 1 \\ & x_3 + x_4 - x_5 & = 1 \\ & -x_3 + x_4 + x_5 & = 1 \\ & -x_2 + x_3 + x_4 & = 1 \end{array}$$

$$\text{Sol. } A_b = \left[\begin{array}{ccccc|c} 1 & 1 & -1 & 0 & 0 & 1 \\ 0 & 1 & 1 & -1 & 0 & 1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 0 & 0 & -1 & 1 & 1 & 1 \\ 0 & -1 & 1 & 1 & 0 & 1 \end{array} \right]$$

$$R \left[\begin{array}{ccccc|c} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 0 & 1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 0 & 0 & -1 & 1 & 1 & 1 \\ 0 & 0 & 2 & 0 & 0 & 2 \end{array} \right], \quad \text{by } R_1 - R_2 \text{ and } R_5 + R_2$$

$$R \left[\begin{array}{ccccc|c} 1 & 0 & 0 & 3 & -2 & 2 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 0 & 0 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & -2 & 2 & 0 \end{array} \right], \quad \begin{array}{l} \text{by } R_1 + 2R_3; R_2 - R_3; \\ R_4 + R_3 \text{ and } R_5 - 2R_3 \end{array}$$

$$\begin{array}{c} R \\ \sim \end{array} \left[\begin{array}{ccccc|c} 1 & 0 & 0 & 3 & -2 & 2 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & -1 & 1 & 0 \end{array} \right], \quad \text{by } \frac{1}{2}R_4 \text{ and } \frac{1}{2}R_5$$

$$\begin{array}{c} R \\ \sim \end{array} \left[\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & -2 & -1 \\ 0 & 1 & 0 & 0 & 1 & 2 \\ 0 & 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{array} \right], \quad \begin{array}{l} \text{by } R_1 - 3R_4, R_2 + 2R_4, \\ R_3 - R_4 \text{ and } R_5 + R_4 \end{array}$$

$$\begin{array}{c} R \\ \sim \end{array} \left[\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{array} \right], \quad \begin{array}{l} \text{by } R_1 + 2R_5, R_2 - R_5, \\ \text{and } R_3 + R_5 \end{array}$$

Clearly $\text{rank } A = \text{rank } A_b = 5$ (the number of unknowns). Hence a unique solution exists and it is

$$x_1 = 1, x_2 = 1, x_3 = 1, x_4 = 1, x_5 = 1.$$

5.
$$\begin{aligned} x_1 - 2x_2 - 7x_3 + 7x_4 &= 5 \\ -x_1 + 2x_2 + 8x_3 - 5x_4 &= -7 \\ 3x_1 - 4x_2 - 17x_3 + 13x_4 &= 14 \\ 2x_1 - 2x_2 - 11x_3 + 8x_4 &= 7 \end{aligned}$$

Sol.
$$A_b = \left[\begin{array}{cccc|c} 1 & -2 & -7 & 7 & 5 \\ -1 & 2 & 8 & -5 & -7 \\ 3 & -4 & -17 & 13 & 14 \\ 2 & -2 & -11 & 8 & 7 \end{array} \right]$$

$$\begin{array}{c} R \\ \sim \end{array} \left[\begin{array}{cccc|c} 1 & -2 & -7 & 7 & 5 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 2 & 4 & -8 & -1 \\ 0 & 2 & 3 & -6 & -3 \end{array} \right], \quad \begin{array}{l} \text{by } R_2 + R_1, R_3 - 3R_1 \\ \text{and } R_4 - 2R_1 \end{array}$$

$$\begin{array}{c} R \\ \sim \end{array} \left[\begin{array}{cccc|c} 1 & 0 & -4 & 1 & 2 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 1 & -2 & 2 \\ 0 & 2 & 3 & -6 & -3 \end{array} \right], \quad \text{by } R_1 + R_4 \text{ and } R_3 - R_4$$

$$\sim R \left[\begin{array}{cccc|c} 1 & 0 & -4 & 1 & 2 \\ 0 & 2 & 3 & -6 & -3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 1 & -2 & 2 \end{array} \right], \text{ by } R_{24}$$

$$\sim R \left[\begin{array}{cccc|c} 1 & 0 & -4 & 1 & 2 \\ 0 & 1 & \frac{3}{2} & -3 & -\frac{3}{2} \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 1 & -2 & 2 \end{array} \right], \text{ by } \frac{1}{2} R_2$$

$$\sim R \left[\begin{array}{cccc|c} 1 & 0 & -4 & 1 & 2 \\ 0 & 1 & \frac{3}{2} & -3 & -\frac{3}{2} \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & -4 & 4 \end{array} \right], \text{ by } R_4 - R_3$$

$$\sim R \left[\begin{array}{cccc|c} 1 & 0 & -4 & 1 & 2 \\ 0 & 1 & \frac{3}{2} & -3 & -\frac{3}{2} \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 1 & -1 \end{array} \right], \text{ by } -\frac{1}{4} R_4$$

We note that $\text{rank } A = \text{rank } A_b = 4$ (the number of unknowns). Hence a unique solution exists and is given by

$$x_1 - 4x_3 + x_4 = 2 \quad (1)$$

$$x_2 - \frac{3}{2}x_3 - 3x_4 = -\frac{3}{2} \quad (2)$$

$$x_3 + 2x_4 = -2 \quad (3)$$

$$x_4 = -1 \quad (4)$$

$$(3) \text{ and } (4) \Rightarrow x_3 - 2 = -2 \Rightarrow x_3 = 0$$

$$(2) \Rightarrow x_2 - \frac{3}{2}(0) - 3 = -\frac{3}{2} \Rightarrow x_2 = \frac{9}{2}$$

$$(1) \Rightarrow x_1 - 4(0) + 1 = 2 \Rightarrow x_1 = 3$$

$$6. \quad x_1 + 2x_2 + x_3 = -1$$

$$6x_1 + x_2 + x_3 = -4$$

$$2x_1 - 3x_2 - x_3 = 0$$

$$x_1 - x_2 = 1$$

$$\begin{aligned}
 \text{Sol. } A_b &= \left[\begin{array}{ccc|c} 1 & 2 & 1 & -1 \\ 6 & 1 & 1 & -4 \\ 2 & -3 & -1 & 0 \\ 1 & -1 & 0 & 1 \end{array} \right] \\
 &\xrightarrow{R} \left[\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 6 & 1 & 1 & -4 \\ 2 & -3 & -1 & 0 \\ 1 & 2 & 1 & -1 \end{array} \right], \text{ by } R_{14} \\
 &\xrightarrow{R} \left[\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 0 & 7 & 1 & -10 \\ 0 & -1 & -1 & -2 \\ 0 & 3 & 1 & -2 \end{array} \right], \text{ by } R_2 - 6R_1, R_3 - 2R_1 \\
 &\quad \text{and } R_4 - R_1 \\
 &\xrightarrow{R} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 0 & -6 & -24 \\ 0 & -1 & -1 & -2 \\ 0 & 0 & -2 & -8 \end{array} \right], \text{ by } R_1 - R_3, R_2 + 7R_3 \\
 &\quad \text{and } R_4 + 3R_3 \\
 &\xrightarrow{R} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & -1 & -1 & -2 \\ 0 & 0 & -6 & -24 \\ 0 & 0 & -2 & -8 \end{array} \right], \text{ by } R_{23} \\
 &\xrightarrow{R} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 1 & 4 \end{array} \right], \text{ by } -1R_2, -\frac{1}{6}R_3 \\
 &\quad \text{and } -\frac{1}{2}R_4 \\
 &\xrightarrow{R} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{array} \right], \text{ by } R_4 - R_3 \\
 &\xrightarrow{R} \left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{array} \right], \text{ by } R_1 - R_3 \text{ and } R_2 - R_3
 \end{aligned}$$

Hence $\text{rank } A = \text{rank } A_b = 3$ (the number of unknowns). Therefore a unique solution exists and it is

$$x_1 = -1, \quad x_2 = -2, \quad x_3 = 4.$$

$$\begin{aligned}
 7. \quad & 2x_1 + x_2 + 5x_3 = 4 \\
 & 3x_1 - 2x_2 + 2x_3 = 2 \\
 & 5x_1 - 8x_2 - 4x_3 = 1
 \end{aligned}$$

$$\text{Sol. } A_b = \left[\begin{array}{ccc|c} 2 & 1 & 5 & 4 \\ 3 & -2 & 2 & 2 \\ 5 & -8 & -4 & 1 \end{array} \right]$$

$$\xrightarrow{R} \left[\begin{array}{ccc|c} 2 & 1 & 5 & 4 \\ 1 & -3 & -3 & -2 \\ 5 & -8 & -4 & 1 \end{array} \right], \quad \text{by } R_2 - R_1$$

$$\xrightarrow{R} \left[\begin{array}{ccc|c} 1 & -3 & -3 & -2 \\ 2 & 1 & 5 & 4 \\ 5 & -8 & -4 & 1 \end{array} \right], \quad \text{by } R_{12}$$

$$\xrightarrow{R} \left[\begin{array}{ccc|c} 1 & -3 & -3 & -2 \\ 0 & 7 & 11 & 8 \\ 0 & 7 & 11 & 11 \end{array} \right], \quad \text{by } R_2 - 2R_1 \text{ and } R_3 - 5R_1$$

$$\xrightarrow{R} \left[\begin{array}{ccc|c} 1 & -3 & -3 & -2 \\ 0 & 7 & 11 & 8 \\ 0 & 0 & 0 & 3 \end{array} \right], \quad \text{by } R_3 - R_2$$

$$\xrightarrow{R} \left[\begin{array}{ccc|c} 1 & -3 & -3 & -2 \\ 0 & 1 & \frac{11}{7} & \frac{8}{7} \\ 0 & 0 & 0 & 1 \end{array} \right], \quad \text{by } \frac{1}{7}R_2 \text{ and } \frac{1}{3}R_3$$

Thus $\text{rank } A = 2$ and $\text{rank } A_b = 3$, so $\text{rank } A \neq \text{rank } A_b$. hence solution of the system does not exist.